



Mythic M2000
Neural Processing Unit (NPU)
Preliminary Datasheet

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Revision History

Updates and changes to this document are reflected in the table below.

Version	Date	Description
0.1	November 2025	Initial version
0.2	February 2026	Minor updates to Sections 3, 6, and 8
0.3	May 2026	Updates to Sections 3, 6, and 8 to add Videantis v-MP integration

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1 Definition of Terms

Abbreviation	Definition	Description
ACE	Analog Compute Engine	Mythic's novel and proprietary hardware engine that performs computation using analog signal processing techniques, enabling high-density and low-power execution of matrix multiplication operations.
ANA8	Analog 8-bit	Analog weights stored in the ACE, providing precision equivalent to 8-bit digital weights.
CIM	Compute-In-Memory	Hardware architecture to perform arithmetic computation directly within memory arrays using analog computation. This reduces data movement bottlenecks and improves performance-per-watt compared to traditional digital compute architectures.
DNN	Deep Neural Network	An artificial neural network with multiple hidden layers positioned between input and output layers, capable of learning hierarchical feature representations and performing complex non-linear inference tasks.
DPU	Dataflow Processing Unit	A processing architecture optimized for dataflow-oriented execution, enabling efficient movement and manipulation of data through a pipeline of compute elements to maximize utilization and throughput.
NPU	Neural Processing Unit	A specialized processor optimized for executing neural network workloads, providing high parallelism and energy-efficient acceleration of machine learning inference.
NPU Tile	Neural Processing Unit Tile	The base component of Mythic's scalable architecture, including ACE, v-MPs, DPU, and SRAM.
SoC	System-on-Chip	An integrated circuit that combines multiple functional components—such as processors, memory interfaces, accelerators, and peripheral controllers—onto a single chip to deliver a complete computing system with reduced power, size, and cost.
v-MP	Videantis - Media Processor	Programmable VLIW/SIMD compute engines for transformers, LLMs, CNNs, vision, DSP, and imaging
v-SP	Videantis - Scalar Processor	Optimized streaming/video codec engines

2 Overview

The Mythic M2000 Neural Processing Unit (NPU) implements a hybrid computing architecture combining unique analog Compute-In-Memory (CIM) technology with digital processing elements to provide high-performance, yet power-efficient, AI inference acceleration. The design of the M2000 uses an array of tiles, each powered by multiple Mythic Analog Compute Engine (Mythic ACE®) instances, a Mythic Dataflow Processing Unit (DPU), and local on-chip SRAM. Each ACE instance contains an array of flash memory, digital-to-analog converters, and analog-to-digital converters. M2000 NPU tiles store weights and provide high-performance, low-power matrix multiplication for inference operations without external DRAM. The M2000 is ideal for processing the complex Deep Neural Networks (DNNs) required for video and sensor fusion analytics in next-generation video surveillance, industrial machine vision, unmanned aerial vehicles, robotics, data center, and consumer electronics applications.

2.1 Features

The M2000 has the following features:

- 12 NPU tiles with the capacity to store up to 60 million total weights
- Up to 370 Tera Operations Per Second (TOPS) of total compute resources
- Mixed precision ANA8 (analog) and INT8/INT16 (digital) computation for performance, accuracy, and power optimization
- Efficient dataflow architecture for deterministic model execution and maximum throughput
- High-speed serial PCIe interface to communicate with a host device
- Die-to-die (D2D) interfaces to scale up to 1,024 M2000 chips in a single system

2.2 Target Applications

The NPU is designed to support a broad range of deep neural network (DNN) workloads. Typical applications include:

- Object detection
- Image classification
- Pose estimation
- Semantic and instance segmentation
- Transformer-based models and LLMs
- Robotics and industrial automation
- Data center acceleration
- Automotive and ADAS systems

3 Functional Description

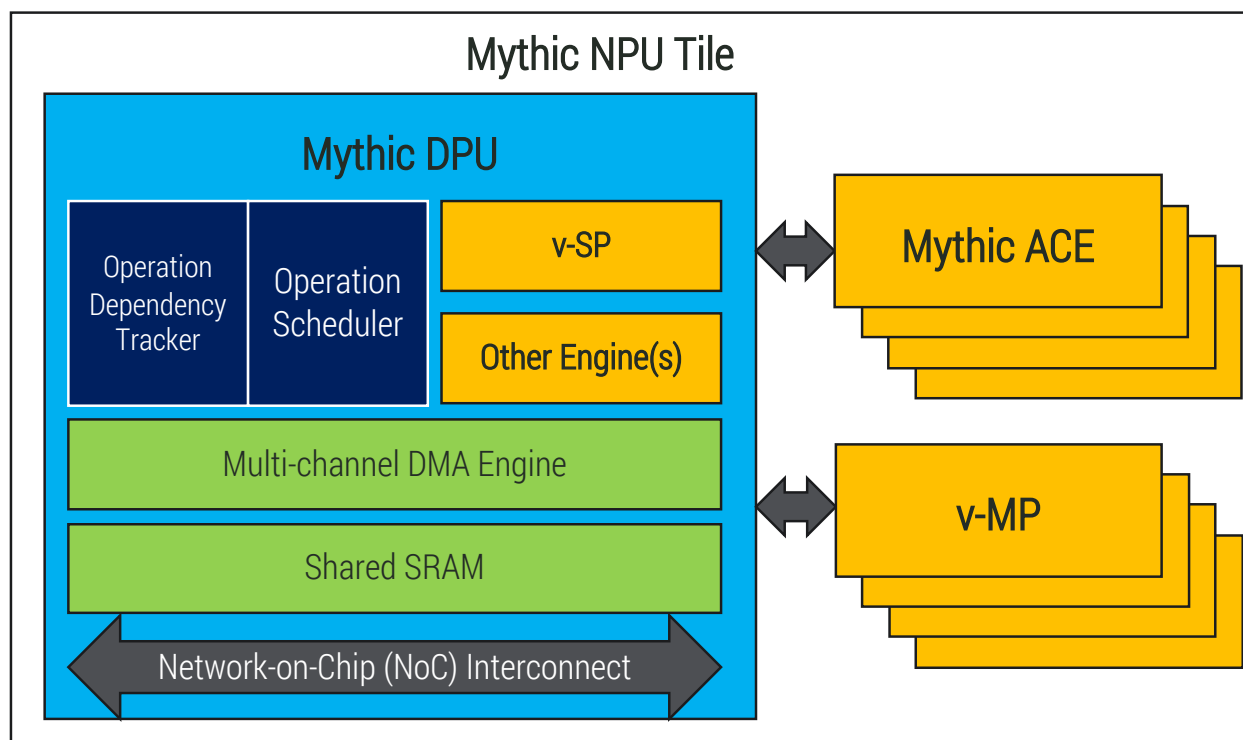
3.1 NPU Architecture

The M2000 has configurable number of NPU tiles on-chip for neural network processing and additional dedicated SoC interface logic to support off-chip communications.

3.1.1 NPU Tile

Each M2000 NPU tile is a hybrid computing system comprised of analog and digital elements as shown in Figure 1.

Figure 1: M2000 NPU Tile



Each NPU tile includes the following components:

- Mythic Analog Compute Engine (Mythic ACE)
 - 1,280 digital inputs
 - 512 Analog-to-Digital Converters (ADCs)
 - 1,280 x 1,024 NVM flash array
- Mythic/Videantis Media Processor Cores (v-MPs)
 - 32-bit scalar data path
 - 64-bit vector data path

- 256-bit wide vector/matrix data path
- Local instruction and data memory
- Mythic Dataflow Processing Unit (DPU)
 - Operation dependency tracker
 - Operation scheduler
 - v-SP, optimized streaming engine
 - Multi-channel DMA engine
 - On-chip SRAM
 - Network-on-chip (NoC) interconnect

3.1.2 SoC Subsystem

The SoC Subsystem includes dedicated logic that enables high-speed communication between the M2000 and a host system via a PCIe interface.

3.2 Mythic Analog Compute Engine (Mythic ACE)

The Mythic ACE uses non-volatile flash memory (NVM) and analog-to-digital converters (ADCs) to perform analog multiply-and-accumulate (MAC) computations for AI applications. Each M2000 NPU tile supports 8-bit digital inputs and outputs. An NVM array is used to store an analog representation of INT8 DNN weights (ANA8) and operates as analog compute elements (multiply inputs with the weights stored in the NVM array). Analog-to-Digital Converters (ADCs) are used to generate 8-bit outputs of the computation by accumulating the results of the flash array compute elements.

3.3 Mythic/Videantis Media Processor Cores (v-MPs)

Each NPU Tile integrates multiple Videantis® v-MP™ programmable media processors to complement Mythic's Analog Compute Engine (ACE) accelerators by efficiently executing the dynamic, irregular, and control-intensive operations that are less suited to fixed-function AI acceleration.

While the ACE architecture delivers industry-leading efficiency for matrix and tensor compute operations used in neural network inference, the v-MP cores provide the programmable flexibility required to support evolving AI models, pre- and post-processing pipelines, and emerging workloads. Together, these engines create a balanced heterogeneous compute platform optimized for real-world AI deployment.

The v-MP architecture is optimized for workloads such as:

- Digital matrix multiplies traditionally seen in transformers and LLMs
- Neural network layers not efficiently mapped to analog tensor acceleration
- Data orchestration and tensor manipulation
- Sparse, conditional, and branching algorithms

- Image signal processing and sensor fusion
- Computer vision and video analytics
- Pre-processing and post-processing for AI inference
- Custom DSP and application-specific acceleration
- Rapid deployment of new and evolving neural network operators

Built on a highly parallel programmable VLIW/SIMD architecture, the v-MP cores provide deterministic real-time performance with exceptional flexibility and performance-per-watt. Their programmable design enables software adaptation to new network topologies, operator types, and application requirements without requiring hardware redesign.

The combination of Mythic ACE acceleration and programmable v-MP processing provides a unified edge AI platform capable of delivering both breakthrough efficiency and long-term algorithmic adaptability.

3.4 Dataflow Processing Unit (DPU)

3.4.1 Theory of Operation

Neural network computation can be represented as a directed acyclic graph (DAG), where nodes correspond to computation units, and arcs denote data dependencies. This abstraction captures the flow of operations and the required ordering of tasks. Such tasks include moving data to and from memory and performing neural network computation.

In Mythic's software and hardware implementations, computation may be divided into finer granularity than an entire layer to optimize the overall performance of the system. An individual computation unit may be configured to process a subset of data as soon as it is available, rather than waiting on data for the full layer at once. Consequently, input dependencies may require only partial results instead of complete layer outputs.

In addition to input data dependencies, computation often requires control dependencies. For example, temporary output buffers or other resources must be available before execution can proceed. These dependencies ensure correct operation and efficient use of memory and processing resources.

A computation unit, therefore, depends on multiple conditions simultaneously, including both valid input data and allocated storage resources. Execution is enabled only when all required dependencies are satisfied, ensuring proper sequencing and resource readiness within the system.

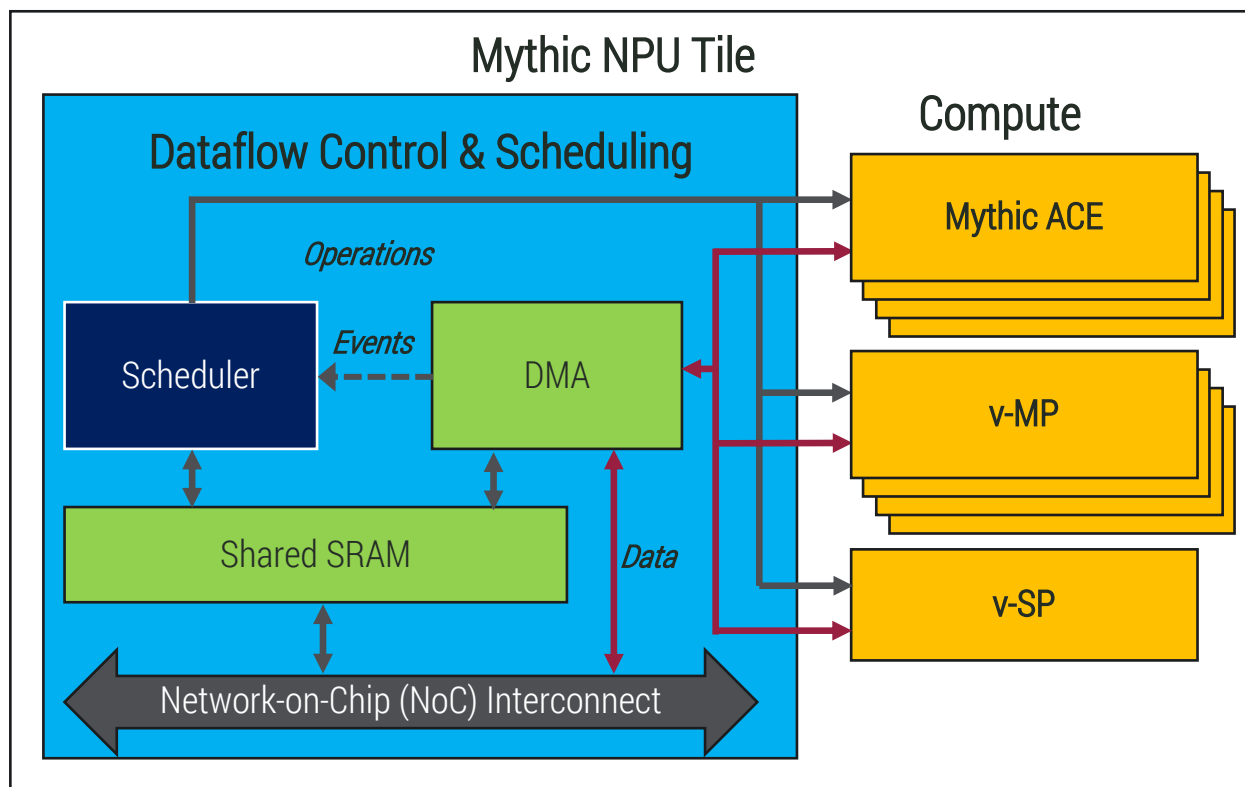
3.4.2 DPU Architecture

The Dataflow Processing Unit (DPU) serves as the digital backbone of Mythic's hardware architecture. It provides a flexible interface to software that enables the Mythic DNN Compiler to efficiently map neural networks onto the hardware while leveraging the Analog Compute Engine (ACE) to accelerate matrix-multiplication operations in the analog domain.

Figure 2 illustrates the architectural overview of the DPU, modified from Figure 1 to emphasize the separation between dataflow control and scheduling elements, and the compute components within the NPU tile, both ACE and v-MP. The scheduler dispatches operations to various hardware blocks, which receive input data from the DMA engines and send output results back to the DMA engines upon completion. When reading from or writing to SRAM buffers, the DMA engines generate event notifications to inform the scheduler, allowing it to issue additional operations to the hardware blocks. The scheduler retrieves operation configurations and related metadata directly from SRAM.

A Network-on-Chip (NoC) interconnect enables each DPU instance to communicate with other DPUs – either on the same chip via an on-chip interconnect, or across multiple chips via the chiplet interface.

Figure 2: Mythic DPU Architectural Overview



3.4.3 DPU Implementation

The M2000 Dataflow Processing Unit (DPU) supports fine-grained computation by representing neural network operations as a network of producers and consumers. The scheduler unit schedules work when all required inputs are available and sufficient resources are available. This event-driven execution model eliminates unnecessary global synchronization and enables high levels of parallelism.

To implement this scheme in hardware, the architecture provides explicit hooks for managing dependencies. Tokens represent data availability, buffer or storage readiness, and other dependencies. The scheduler processes tokens and marks appropriate operations as eligible for scheduling once all the operation's dependencies have been met. This mechanism ensures correctness while allowing the hardware to leverage available concurrency.

Control and scheduling logic are distributed across the architecture rather than centralized. Each NPU tile maintains its operation dependencies locally, reducing global coordination overhead. The result is a scalable execution model that can adapt to workloads with varying granularity, from pixel-level operations to full-layer processing.

By combining fine-grained dependency management with distributed scheduling, the Mythic DPU architecture provides the necessary foundation for efficient neural network execution. It ensures that

computation units launch only when data and resources are ready, enabling predictable performance and high hardware utilization.

3.4.4 DPU Hardware Features

3.4.4.1 *Local Dependency Tracking*

The dependency tracking unit contains logic to track input data tokens and resource tokens. This allows units to self-determine readiness, eliminating reliance on centralized scheduling.

3.4.4.2 *Distributed Scheduling*

Execution is event-driven and managed locally in each NPU tile. Hardware operations are scheduled when dependencies are met, enabling fine-grained concurrency and reducing idle cycles caused by global synchronization.

3.4.4.3 *Buffer Management*

Temporary storage resources are managed dynamically. Buffer tokens ensure that output space is allocated before operation execution begins, minimizing bottlenecks, and preventing resource conflicts.

3.4.4.4 *Arbitration and Flow Control*

Data and control tokens are routed through low-latency interconnects. Arbitration logic ensures fair access to shared resources while maintaining functionally correct execution ordering.

3.4.4.5 *Scalability*

The architecture scales from small, low-power implementations to wide, high-throughput arrays of compute units. Granularity can be tuned to match workload requirements, from pixel-level operations to layer-level processing.

3.4.4.6 *Predictable Performance*

By enforcing execution only when both data and resources are available, the system ensures predictable latency, high utilization, and efficient throughput across varying workloads.

3.5 Primary NPU Interface

PCIe serves as the high-bandwidth data interface to an external host processor. The PCIe interface comprises physical, link, and transaction layers. It also implements Mythic-specific DMA logic for host driver and tile SRAM data transfers.

The PCIe interface is used for the following operations:

- Transfer DNN weights to M2000
- Load firmware
- Transfer input data from the host to M2000 and inference results from M2000 to the host
- Provide status and performance data to the host

3.6 SIMD Engine

Each NPU tile contains a digital SIMD engine to perform arithmetic operations that are better optimized for execution outside the ACE. These include graph nodes requiring higher precision than the ACE provides, as well as operations such as pooling and digital matrix multiplies. By distributing the SIMD engines throughout the digital architecture, software can take advantage of localized analog and digital computation. This reduces the need to transfer large volumes of data between NPU tiles while processing successive layers of the neural network.

3.7 Network-on-Chip (NoC)

The Network-on-Chip (NoC) serves as the communication backbone for the device, enabling data transfer between NPU tiles. Within each tile, the Dataflow Processing Unit (DPU) components connect to the NoC for local communication. At the same time, the DPU interfaces with the global NoC fabric to exchange data and control flow transactions with DPUs on other tiles. This dual-level connectivity supports both fine-grained intra-tile coordination and large-scale inter-tile transfers, facilitating efficient execution of distributed neural network workloads.

3.8 Die-to-die (D2D) Interface

The homogeneous die-to-die interface on the M2000 provides scalable connectivity in all four cardinal directions, enabling seamless expansion of the architecture. Within a package, the interface supports short-distance links between adjacent dies, offering high bandwidth and low latency. For extended scalability, the same interface operates across packages, supporting cross-package connections while maintaining protocol consistency. This unified approach simplifies system integration, allowing designers to scale performance linearly by adding dies in-package or across multiple packages.

4 M2000 NPU Software Toolkit

Mythic provides a software toolkit with the M2000 NPU. The software toolkit workflow enables users to convert, optimize, and compile an industry standard DNN for execution on the M2000. The software toolkit also supports retraining of the DNN to optimize the weights for a given user dataset without changing the structure of the DNN itself. After successful conversion, optimization, and retraining of the DNN, the workflow enables users to compile the DNN and weights into a set of binaries for execution on the M2000.

The M2000 NPU Software Toolkit also includes the necessary host-based drivers and utilities required to initialize the M2000 for execution of a specific DNN and to debug and optimize runtime performance.

4.1 DNN Conversion and Optimization

The M2000 NPU Software Toolkit includes a conversion tool to convert DNNs designed in industry standard frameworks such as PyTorch for inference operation on the M2000. The conversion tool modifies each DNN layer and operators for execution in M2000 hardware blocks. The DNN may then be optimized for performance for customer applications.

4.2 Graph Compiler

The M2000 NPU Software Toolkit includes a graph compiler that enables users to convert a DNN, defined in ONNX, into binaries that are executed on the M2000.

4.3 Binaries

The M2000 NPU graph compiler creates a set of binaries that includes:

- The weights for each NPU tile used to execute the DNN on the M2000
- A binary image of the firmware for each NPU tile

The binary format is proprietary and not user modifiable. Mythic does not guarantee that the M2000 will execute correctly if the binaries are modified outside of the M2000 NPU software toolkit.

4.4 Drivers and Utilities

The M2000 NPU software toolkit provides a set of Linux-based host drivers which control communication between the host processor and the M2000. The software toolkit also includes utilities that can be used to debug and analyze the operation of the M2000 while running inference. These drivers and utilities allow users to control the M2000 in their system through both Command Line Interfaces (CLIs) and Application Programming Interfaces (APIs).

Upon boot, the M2000 will establish communication with the host processor and will then load firmware for each NPU tile. The M2000 runtime software manages programming of DNN weights as part of the chip initialization sequence. Once this initialization is complete, the M2000 will wait for the host processor to supply data to be processed for inference.

5 NPU Interfaces

5.1 Host Interface

The M2000 NPU is a standalone chiplet that connects to the host device using a Gen4 PCIe interface.

5.2 Die-to-Die (D2D) Interfaces

A system may contain multiple instances of the M2000 NPU chiplet. For communication between the chiplets, a die-to-die interface is used. This interface is UCIe and transports AXI4 transactions over the interface to connected chiplets.

The NPU contains four D2D interfaces extending the communication in all four cardinal directions – North, South, East, and West.

6 Performance and Power Targets

Table 1 lists the estimated M2000 performance and power consumption for popular neural network applications during inference. The device also supports a standby mode; standby power consumption will be provided in a future revision of this datasheet.

Inference power can vary depending on the end-user application and factors such as DNN model size, inference latency (performance), and temperature. The values in Table 2 are estimated under nominal operating conditions. The device ambient operating temperature range is -40°C to 125°C .

Table 1: Performance and Power Targets

Application	Input Frame Resolution (Pixels)	Estimated Inference Rate (fps)	Estimated Power (W)
ResNet-50	224x224	1830	1.1
YOLOv8s-Object Detection	1088x1920	30	0.3
		57	0.6
YOLOv8s-Pose Detection	1088x1920	30	0.4
		45	0.6
YOLOPX	736x1280	30	0.7
		106	2.2
BEVFormer-Tiny	6x928x1600	30	4.4

7 M2000 NPU Tile Flash Memory Arrays

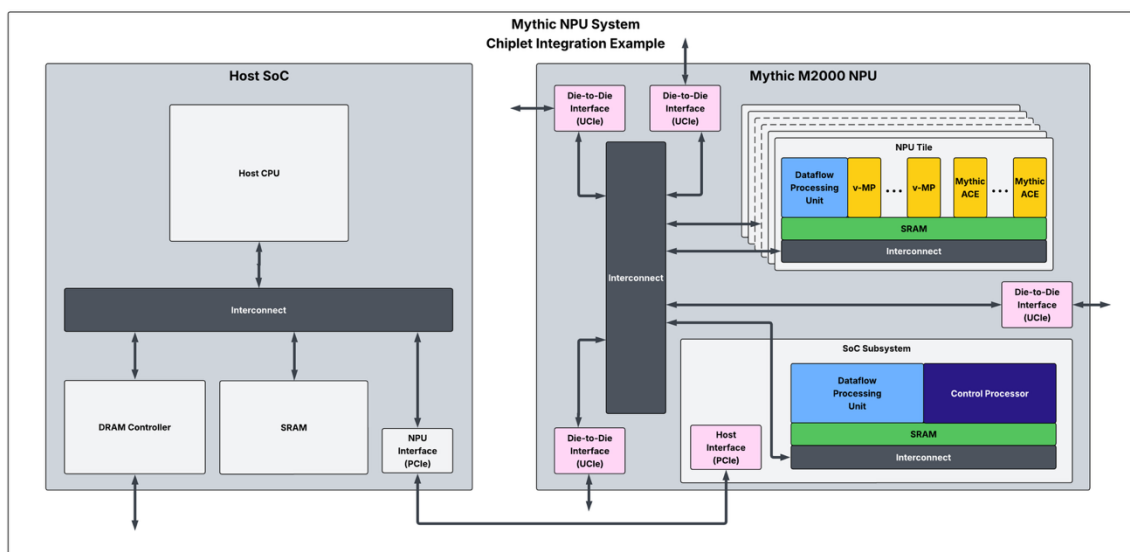
Prior to running inference on the M2000, the DNN weights must first be programmed into the NPU tile flash memory arrays. Compiled binaries for the M2000 contain the necessary weights and their respective tile mappings. The weights are programmed only during DNN initialization (i.e., offline). Weights do not need to be reprogrammed between different inference runs, unless the DNN needs to be replaced or updated.

8 IP and System Integration

8.1 Chiplet System Integration

The M2000 NPU is delivered as a standalone chiplet and is not intended for direct integration as an IP block within a monolithic SoC. The integrated PCIe interface enables high-bandwidth data transfers between the host SoC and the M2000 NPU chiplet for neural network processing as shown in Figure 3.

Figure 3: M2000 Chiplet Integration Example



8.2 Mythic Runtime Software

Future revisions of this document will describe the Mythic Runtime Software, which manages communication between the M2000 NPU and the host system. Topics to be covered include boot procedures, flash programming, error detection and reporting, fault recovery, and debugging capabilities.